

Malware

Definition: Malware, or malicious software, is any malicious program or code that is harmful to systems. It is designed to invade, damage, disable, spy on, or otherwise interfere with computers, systems, networks, and devices.

What Malware Can Do

- Steal sensitive information.
- Encrypt or delete data.
- Alter or hijack core computer functions.
- Monitor computer activity without the user's knowledge or permission.
- Interfere with normal system functioning.
- In some industrial environments, manipulate hardware operations and potentially cause physical damage.

Motives Behind Malware

The motives can vary. Malware may be used to make money, sabotage a victim's ability to work, make a political statement, or demonstrate an attacker's capabilities or status.

Important: Malware typically does not directly damage physical hardware. However, in rare cases—especially in industrial systems—malware can cause physical consequences by manipulating how hardware operates.

1. Signs of Malware

Possible signs of malware infection include:

- Computer becomes unusually slow.
- Screen is inundated with annoying or unexpected advertisements and pop-ups.
- System freezes, crashes, or displays a Blue Screen of Death (BSOD).
- Mysterious loss of disk space.
- Unexpected increase in internet or network usage.
- Browser settings change without the user's intention.
- Loss of access to system files.

Note: These symptoms do not prove that malware is present. Some malware can remain deep inside a system and operate without obvious warning signs.

2. Types of Malware

Type	Description	Example / Note
Adware	Unwanted software designed to display advertisements, often through a web browser. It may disguise itself as legitimate software or be bundled with another program.	Browser hijackers
Spyware	Malware that secretly observes a user's activities without permission and reports information to its author.	Secret monitoring of user activity
Virus	Malware that attaches to another program and replicates itself by modifying other programs when the infected program is executed.	ILOVEYOU virus
Worm	Self-replicating malware that can spread across systems on its own, without requiring the same user action associated with a traditional virus.	WannaCry; SQL Slammer
Trojan Horse	Malware that presents itself as something useful or legitimate to trick users into installing it. It can provide unauthorized access or install additional malware.	Fake cracked software; Zeus Trojan
Ransomware	Malware that can lock users out of devices and/or encrypt files and demand a ransom to regain access.	WannaCry; Locky; Ryuk
Rootkit	Malware designed to provide privileged/root access and remain hidden from users, software, or the operating system.	Sony BMG rootkit incident
Keylogger	Malware that records keystrokes, potentially capturing usernames, passwords, payment information, and other sensitive data.	Capturing typed credentials
Cryptojacking	Malicious cryptomining that uses another person's computing resources to mine cryptocurrency for the attacker's benefit.	Unauthorized cryptocurrency mining

3. Adware

Adware is unwanted software designed to display advertisements, most often within a web browser. It may use deceptive techniques to appear legitimate or may be bundled with another program, tricking the user into installing it.

Example: Browser hijackers.

4. Spyware

Spyware secretly observes a computer user's activities without permission and reports the collected information to the software's author.

5. Virus

A virus is malware that attaches itself to another program. When the infected program is executed, the virus can replicate by modifying other programs and inserting its own code.

Example: ILOVEYOU virus.

6. Worm

Worms are self-replicating malware. A key distinction from traditional viruses is that worms can spread from system to system on their own, whereas viruses generally depend on some form of user action or execution of an infected host program.

Example: WannaCry combined ransomware behavior with worm-like propagation. SQL Slammer is another example of rapidly spreading worm malware.

7. Trojan Horse

A Trojan, or Trojan horse, presents itself as something useful or legitimate to trick a user into installing it. Once installed, it can provide unauthorized access, steal information, or install additional malware. Trojans can also be used to establish a backdoor for attackers.

Examples: Fake cracked software and Zeus Trojan.

8. Ransomware

Ransomware is malware that can lock a user out of a device and/or encrypt files, then demand payment in exchange for restoring access. It can target both individuals and organizations.

Examples: WannaCry, Locky, and Ryuk. Ryuk was described in the source material as targeting high-profile organizations that could be more likely to pay large ransoms.

9. Rootkit

A rootkit is a form of malware associated with obtaining administrator or root-level privileges on an infected system and maintaining a high degree of stealth. It may be designed to hide from users, applications, and operating-system components.

Example: Sony BMG rootkit incident.

10. Keylogger

A keylogger records a user's keystrokes. The captured information may be stored locally or sent to an attacker, who may seek usernames, passwords, payment details, or other sensitive information.

11. Cryptojacking

Cryptojacking is the unauthorized use of another person's computing resources for cryptocurrency mining. The attacker benefits from the victim's CPU, GPU, electricity, and system resources rather than using their own resources.

12. Malware Mitigation

The source material recommends the following practices to reduce malware risk:

- **1.** Pay attention to the domain and be cautious when a website appears suspicious or unfamiliar.
- **2.** Use strong passwords with multi-factor authentication. A password manager can help manage unique passwords.
- **3.** Avoid clicking on pop-up advertisements while browsing the internet.
- **4.** Avoid opening email attachments from unknown senders.
- **5.** Do not click strange or unverified links in emails, texts, or social-media messages.
- **6.** Do not download software from untrustworthy websites or peer-to-peer file-transfer networks.
- **7.** Prefer official app stores and trusted software sources. Avoid jailbreaking mobile devices and check software reputation before installation.
- **8.** Keep the operating system, browsers, plugins, and other software patched and up to date.
- **9.** Remove programs that are no longer needed.
- **10.** Back up important data regularly so that files can be recovered if they become damaged, encrypted, or inaccessible.

Key Takeaway

Malware is a broad category of malicious software. Different malware types use different methods and have different objectives, but common outcomes include unauthorized access, data theft, disruption, surveillance, resource abuse, and data encryption or destruction. Good security hygiene—especially patching, cautious handling of links and attachments, strong authentication, trusted software sources, and regular backups—helps reduce the risk and impact of malware.